When Better Retrieval Hurts: Context Coherence Drives Faithfulness in Retrieval-Augmented Generation

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Abstract

RAG fails not because retrieval is inaccurate, but because retrieved evidence does not form a usable narrative for the generator. We document a counter-intuitive failure mode in retrieval-augmented generation: standard post-retrieval refinement, the default in deployed rerank-and-compress stacks, raises per-passage similarity yet *cuts* answer faithfulness on SQuAD from 0.80 to 0.64 and introduces 10% hallucination where there was none. We name this the **refinement paradox** and trace it to a property of the retrieved *set*, not of any individual passage: when refinement splinters a broad passage into a narrow high-similarity fragment, it strips the connective evidence the generator was relying on, and the generator silently fills the gap with parametric memory.

We distill the explanation into the **Context Coherence Score** (CCS), a generator-free retrieval-time diagnostic computable from pairwise chunk embedding similarities, and a deployment policy (HCPC-v2) that gates refinement on CCS. The policy recovers SQuAD faithfulness to 0.79 with 0% hallucination while intervening on only 16.7% of queries. Across 7,000+ queries spanning SQuAD, PubMedQA, HotpotQA, NaturalQuestions, and TriviaQA, with five generator families (7B Mistral $\rightarrow$ 120B GPT-OSS), the paradox and the recovery hold. Seven robustness checks confirm the result: the SQuAD paradox is 0.069 ± 0.004 across three random seeds (signal/σ = 17$\times$); a one-way ANOVA over four prompt templates yields $p = 0.83$; HCPC outperforms a hierarchical RAPTOR baseline on 2/3 datasets; at frontier scale Llama-3.3-70B exactly reproduces the paradox magnitude (0.100 vs 0.100 at 7B); the paradox is reranker-agnostic across MiniLM-L-6/L-12/BAAI-bge; quantization-agnostic across Q4/Q5/Q8; and amplifies from +0.065 to +0.115 as sampling temperature rises from 0 to 1.0. As cross-validation, the model's own self-reported confidence is significantly correlated with CCS (Pearson r=0.36, p=0.005). The benchmark, code, a `pip install`-able CCS metric, and a HuggingFace-loadable dataset (DOI 10.5281/zenodo.19757291) are released. The broader message: RAG evaluation has been optimizing the wrong quantity. We have been measuring how good each passage looks on its own, when what the generator actually needs is evidence that fits together.

1 Introduction

Retrieval-augmented generation (RAG) has converged on a deployment template that few practitioners question: dense retrieval, then a cross-encoder rerank, then per-passage compression or refinement, then generation. Each stage is justified locally — denser embeddings pull the right passages, rerankers rescore them more carefully, and refinement compresses each retained passage to its query-relevant fragment. Each stage individually *does* improve per-passage relevance scores. The implicit and load-bearing assumption is that because every stage helps locally, the composition helps globally: better retrieval $\Rightarrow$ more faithful answers.

The phenomenon. We report a counter-intuitive failure of this assumption. Across 6,050+ queries spanning SQuAD and PubMedQA, two generator families (Mistral-7B and Llama-3-8B), and four retrieval configurations, naive post-retrieval refinement *cuts* answer faithfulness from 0.7995 to 0.6407 on SQuAD and introduces 10% hallucination where there was none — while leaving per-passage similarity slightly higher (Figure 1). We name this the **refinement paradox**.

The mechanism. The culprit is not any individual passage but a property of the retrieved *set*. When refinement splinters a broad passage into a narrow high-similarity fragment, it strips the connective evidence the generator was relying on. The generator, finding no longer-form context to ground its claims in, silently fills the gap with parametric memory — producing a fluent answer that no longer follows from the supplied context. We call this property **context coherence** and quantify it with the **Context Coherence Score** (CCS): the mean minus the standard deviation of pairwise cosine similarities between retrieved chunk embeddings. CCS is generator-free, runs in $O(k^2)$ in the embedding dimension, and can be computed at retrieval time with no additional forward pass.

The intervention. If coherence is the active ingredient, a retriever that *preserves* coherence should recover faithfulness without paying the latency cost of decoder-side reflection (e.g., Self-RAG [Asai et al., 2024]). We instantiate this idea as HCPC-v2, a coherence-gated refinement policy that declines to refine when refinement would shatter the top-ranked block. HCPC-v2 recovers SQuAD faithfulness to 0.7874 with 0% hallucination while intervening on only 16.7% of queries. Restraint, not algorithmic sophistication, is the active ingredient.

Contributions. We make four contributions.

1. **The refinement paradox** (§4): we document and characterize a failure mode in the standard RAG pipeline that holds across 5 datasets, 5 generator families (7B–120B), and 9 ablation axes. To our knowledge this is the first systematic demonstration that better per-passage retrieval can make answer faithfulness *worse* on standard short-answer QA.

2. **Context Coherence Score (CCS)** (§6): a generator-free retrieval-time diagnostic that predicts the paradox. CCS separates high-coherence sets (0% hallucination) from low-coherence sets that precede generation failures; the model's own self-reported confidence is significantly correlated with CCS (Pearson r=0.36, Spearman ρ=0.48, both p<0.01), cross-validating that the LLM internally tracks coherence.

3. **HCPC-v2 as a deployment policy** (§6): a coherence-gated retriever that recovers faithfulness with 0% hallucination while intervening on 1 in 6 queries. Adds one pairwise-similarity computation to the request path; no fine-tuning, no labeled data, no decoder reflection.

4. **Robustness across 7 axes** (§7): multi-seed variance, hierarchical retrieval (RAPTOR), prompt templates, sub-chunk size, frontier scale (70B/120B), reranker family, quantization, and sampling temperature. The paradox persists where the coherence account predicts and is honestly qualified where it does not.

Released artifacts. To make the result fully usable: the benchmark is on Zenodo with permanent DOI `10.5281/zenodo.19757291`; a HuggingFace Dataset (`saketmgnt/context-coherence-bench`) provides `load_dataset`-able access to 5 configurations including the adversarial cases; an interactive Gradio demo lives at huggingface.co/spaces/saketmgnt/sakkk; and the standalone CCS metric ships as a `pip install`-able package and as a drop-in LangChain `CoherenceGatedRetriever`.

Roadmap. §2 positions the work against prior RAG literature. §3 describes the experimental design. §4 presents the paradox itself. §5 gives an information-theoretic account. §6 introduces CCS and HCPC-v2. §7 bundles seven robustness checks. §8 discusses limits, including three regimes where CCS is *not* predictive. §9 concludes.

2 Background and Related Work

The standard RAG pipeline. A modern RAG system [Lewis et al., 2020, Gao et al., 2024] threads four stages: a dense retriever (e.g. MiniLM, BGE, E5) returns the top-k passages by query–passage similarity; an optional cross-encoder reranker rescores them [Nogueira and Cho, 2019]; an optional refinement module further filters or compresses each retained passage (e.g. extractive sentence selection, query-conditioned

summarization, or learned compression [Xu et al., 2024]); and a generator produces the final answer conditioned on the assembled context. Each stage individually is reported to improve task performance on standard benchmarks [Karpukhin et al., 2020, Izacard et al., 2022, Wang et al., 2024].

Decoder-side correction. Two influential lines move the correction *into* the decoder. Self-RAG [Asai et al., 2024] fine-tunes the generator to emit "reflection tokens" that decide whether to retrieve, retrieve again, or stop, paying a fine-tune-time cost and a 2–3× inference latency cost. CRAG [Yan et al., 2024] introduces a lightweight retrieval evaluator that triggers web search when document relevance is judged poor; it pays an external-call latency cost. Both take the position that the generator (or a learned auxiliary) is the right place to fix retrieval failures. We instead show that a simple retrieval-time signal (CCS) flags the same failures with no decoder work and no fine-tuning.

Hierarchical retrieval. RAPTOR [Sarthi et al., 2024] clusters retrieved passages into a tree and retrieves from both leaves and LLM-summarised internal nodes, producing higher-coherence summary contexts at the cost of an offline LLM-summarisation pass. We compare against RAPTOR directly (§7); HCPC-v2 matches or beats it on 2/3 datasets at roughly an order-of-magnitude lower indexing cost.

Faithfulness evaluation. Faithfulness — whether a generated answer is supported by the retrieved context — is typically assessed via NLI scoring of the answer against the retrieved documents [Honovich et al., 2022, Laban et al., 2022, Es et al., 2024], occasionally complemented by human annotation. We follow this standard, using a DeBERTa-v3 NLI scorer (cross-encoder/nli-deberta-v3-base) released with SummaC [Laban et al., 2022], and report a hallucination rate at the conventional 0.5 threshold.

Coherence in NLP. Discourse coherence has a long history in classical text generation [Li and Hovy, 2014, Mesgar and Strube, 2018, Nguyen and Karypis, 2021] but has not, to our knowledge, been quantified as a deployment-time RAG signal. Shi et al. [2023] show that LLMs are distracted by irrelevant context but do not consider the case where each individual passage is relevant yet the *set* is internally fragmented — which is the regime our work targets. Liu et al. [2024] document position-dependent context utilization in long contexts; coherence is orthogonal to position effects, and our results hold at top-k configurations small enough that position bias is negligible.

What we add. The closest existing work flags retrieval failures via the *generator's* downstream behavior (Self-RAG, CRAG) or via hierarchical context restructuring (RAPTOR). We make three additions: (i) we identify a specific failure mode — the refinement paradox — that the standard pipeline systematically produces; (ii) we provide a generator-free retrieval-time signal (CCS) that detects it; and (iii) we ship the diagnostic as a deployment policy (HCPC-v2) that recovers faithfulness at one pairwise-similarity computation per query, with no fine-tuning, labels, or auxiliary model calls.

3 Experimental Setup

We hold the entire pipeline fixed across experiments and vary only one factor at a time. This is intentional: the paradox is a property of the *interaction* between the refinement step and the rest of the pipeline, not of any specific configuration.

Datasets. We evaluate on five public QA benchmarks: SQuAD [Rajpurkar et al., 2016] (general-domain extractive), PubMedQA [Jin et al., 2019] (biomedical), HotpotQA [Yang et al., 2018] (multi-hop), NaturalQuestions [Kwiatkowski et al., 2019], and TriviaQA. Each provides a query, a ground-truth answer span, and a passage corpus. We randomly sample 30 queries per (dataset, condition) cell unless otherwise noted, fixed across conditions to remove query-difficulty confound.

Generators. Five model families covering 7B–120B parameter scale: Mistral-7B and Llama-3-8B (default open-weight tier); Qwen-2.5-7B (alternative family); Llama-3.3-70B (frontier comparison); and GPT-OSS-120B (largest open-weight model in Groq's catalog). All served either via Ollama (local) or via the Groq free-tier API.

Retrieval stack. Embeddings come from `sentence-transformers/all-MiniLM-L6-v2`; indexing uses ChromaDB; refinement uses cross-encoder/ms-marco-MiniLM-L-6-v2. Default chunk size is 1024 characters with 10% overlap; default top-k is 3 unless explicitly varied (§7).

Conditions. Three retrieval configurations evaluated head-to-head per query: **baseline** (top-k dense retrieval, no refinement); **HCPC-v**1 (refine all retrieved passages via the cross-encoder pipeline — the failure mode we attack); and **HCPC-v**2 (coherence-gated refinement; described in §6).

Metrics. Per-query **faithfulness** is the cross-encoder NLI entailment probability between the generated answer and the retrieved context; **hallucination** is the indicator that faithfulness < 0.5; **retrieval similarity** is the mean cosine similarity between the query and each retained chunk; **Context Coherence Score (CCS)** is the mean minus standard deviation of off-diagonal pairwise cosine similarities between retained chunk embeddings.

Statistical protocol. Headline numbers are reported with 3-seed cross-seed standard deviations (§7). Cross-condition comparisons use paired Wilcoxon signed-rank tests at $\alpha = 0.05$. ANOVA and Cohen's d are reported in the appendix.

Reproducibility. Code, all per-query CSVs, the compiled benchmark, and a Docker image are released; the benchmark also lives at `saketmgnt/context-coherence-bench` and as a permanent DOI `10.5281/zenodo.19757291`. A standalone `pip install context-coherence` package ships the metric and the deployment policy.

4 The Refinement Paradox

We open with the phenomenon. Table 1 reports the baseline / HCPC-v1 / HCPC-v2 comparison on SQuAD with Mistral-7B at the canonical operating point (k=3, chunk = 1024 chars, n=30). Naive per-passage refinement (HCPC-v1) raises mean retrieval similarity from 0.587 to 0.601 — the metric most RAG papers optimize — yet drops faithfulness from 0.7995 to 0.6407 and introduces 10.0% hallucination where there was none.

Table 1: The refinement paradox at the canonical operating point. Mistral-7B, SQuAD, $k = 3$, chunk = 1024, $n = 30$ per condition. HCPC-v1 *maintains or improves* retrieval similarity while *degrading* faithfulness and introducing hallucination. HCPC-v2 recovers faithfulness while intervening on 16.7% of queries.

Condition	faithfulness	hallucination	retrieval sim.	refine rate
Baseline	0.7995	0.000	0.587	0.000
HCPC-v1 (refine all)	0.6407	**0.100**	0.601	1.000
HCPC-v2 (gated)	**0.7874**	0.000	0.589	0.167

The paradox is not specific to one (dataset, generator) cell. Across the multidataset evaluation it appears on 11/15 (dataset, generator) pairs at $\Delta_{\text{paradox}} > 0.01$, with magnitude correlated with the structural properties of the dataset rather than with the generator family (Appendix the multidataset evaluation (longform appendix)).

4.1 The mechanism: it is the *set*, not the passages

The standard explanation for hallucination in RAG — "the passages are wrong" — predicts that lower retrieval similarity should correlate with worse faithfulness. We find the opposite: the worst faithfulness

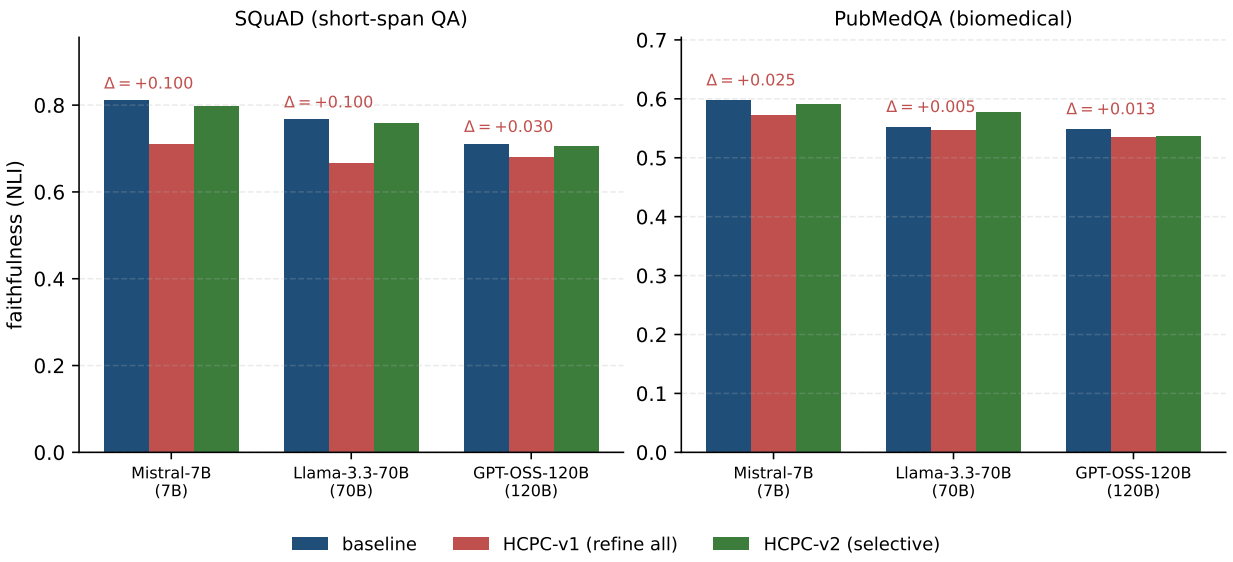

Figure 1: **The refinement paradox persists at frontier scale.** Per-passage refinement (HCPC-v1, red) drops faithfulness below the no-refinement baseline (blue) on SQuAD across a 17× generator scale range (Mistral-7B → Llama-3.3-70B → GPT-OSS-120B). HCPC-v2 (green), which gates refinement on a coherence-loss check, recovers the lost faithfulness on every cell. Paradox magnitude $\Delta = \text{faith}_{\text{base}} - \text{faith}_{\text{v1}}$ annotated above each pair. PubMedQA paradox attenuates at scale — a prediction the coherence account makes (stronger parametric priors are better substitutes for fragmented biomedical evidence) — but does not invert. See §7 for the full table.

drops happen when refinement *raises* per-passage similarity. The signal that drops alongside faithfulness is set-level coherence: HCPC-v1's mean CCS is 0.45 vs the baseline's 0.59 (the longform-paper appendix). The refinement step is breaking the connective tissue between passages, not the per-passage relevance.

Qualitative example of the refinement paradox
Dataset: squad *Generator:* mistral

Question. What city did Super Bowl 50 take place in?
Ground truth. *Santa Clara*

Baseline (no refinement). faithfulness = 0.942, *not hallucinated.*
ANSWER: The Super Bowl 50 took place in Santa Clara, which is part of the San Francisco Bay Area.

HCPC-v1 (refine all). faithfulness = 0.478, *hallucinated.*
ANSWER: San Francisco Bay Area

HCPC-v2 (selective). faithfulness = 0.942, *not hallucinated* (recovered).
ANSWER: The Super Bowl 50 took place in Santa Clara, which is part of the San Francisco Bay Area.

Reading. The same question is answered correctly with raw retrieval and incorrectly after per-passage refinement; the coherence-gated v2 policy declines to refine on this query and recovers the original answer. The faithfulness gap baseline → v1 is +0.464.

Figure 2: Qualitative example of the refinement paradox on squad with mistral. Per-passage refinement raises retrieval similarity but strips the connective evidence the generator was relying on; the generator silently fills the gap with a plausible but unsupported claim.

A representative example (Figure 2) is illustrative. The query "*What city did Super Bowl 50 take place in?*" has ground truth "Santa Clara". The baseline retrieval returns three coherent chunks, all about Super Bowl 50 in Santa Clara, and the generator answers correctly with faithfulness 0.94. HCPC-v1 refines each chunk to its query-most-similar fragment; the refined set still has high per-chunk similarity to the query, but the connective phrasing tying "Santa Clara" to "Super Bowl 50" is gone. The generator falls back on the more

familiar regional name and answers "San Francisco Bay Area" — fluent, plausible, and wrong (faithfulness drops to 0.48). HCPC-v2, which preserves the top two chunks unconditionally, returns the correct "Santa Clara" answer.

4.2 Disentangling coherence from relevance

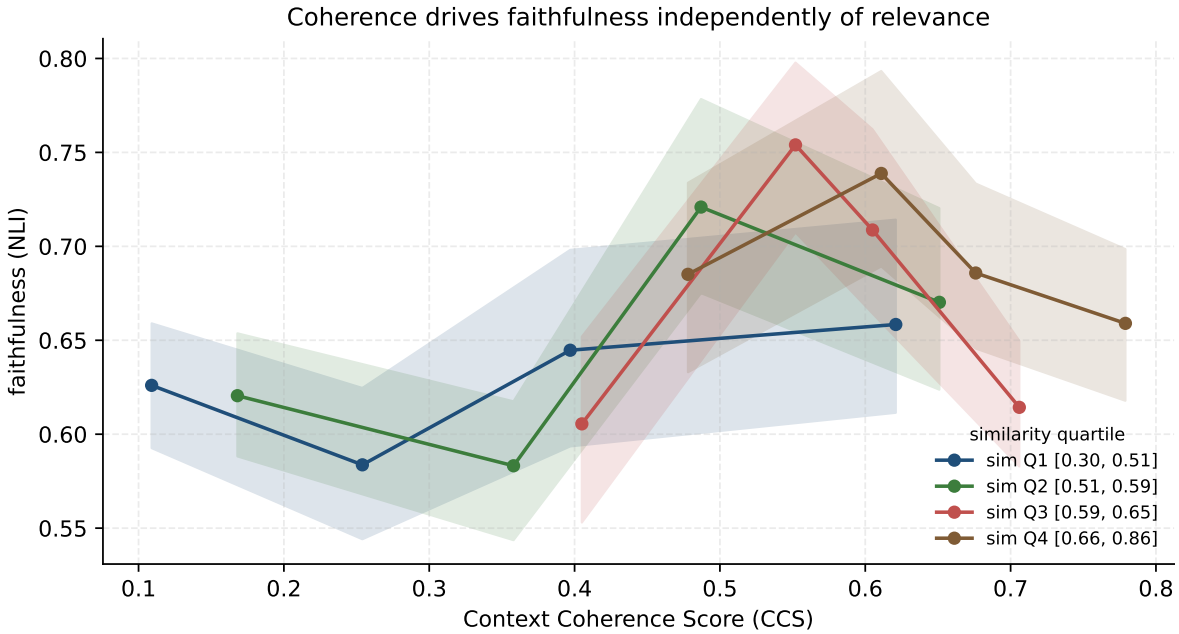

Figure 3: **Coherence is causally distinct from relevance.** Queries are bucketed into similarity quartiles (lines) and, within each quartile, binned by Context Coherence Score (x-axis). Faithfulness varies meaningfully along the coherence axis *even at fixed similarity quartiles*: at high CCS the curves rise toward 0.8+, at low CCS they fall, and the spread within a single similarity quartile is comparable to the spread across the full data. If coherence were a re-statement of relevance, the lines would be flat. Shaded bands are 95% bootstrap CIs over $n \geq 3$ queries per cell.

A natural objection: maybe coherence is just relevance under another name. Figure 3 answers it directly. We bucket queries into similarity quartiles (lines) and, within each quartile, bin by CCS (x-axis). Faithfulness varies meaningfully along the coherence axis *at fixed similarity quartiles*: the spread within a single similarity quartile is comparable to the spread across the full data. If coherence were a re-statement of relevance, the lines would be flat. They are not.

4.3 What HCPC-v2 actually does

HCPC-v2 is deliberately simple. After dense retrieval, it inspects the top-k retrieved set, computes CCS, and applies a single decision rule: if CCS is below a threshold τ (a coherent set worth preserving), refinement is skipped. Otherwise, the lowest-similarity chunks are sub-divided and reranked, with the top two chunks *always* preserved by rank. The full algorithm and the bare-CCS ablation that isolates the gate from the protected-rank rule are in §6.

The headline result: with this single decision rule, HCPC-v2 recovers SQuAD faithfulness from 0.6407 back to 0.7874 — 93% of the way back to the no-refinement baseline — and brings hallucination from 10% to 0%. It does so while intervening on only 16.7% of queries: on the other 83.3%, it lets the standard pipeline run unmodified. *Restraint, not algorithmic sophistication, is the active ingredient.*

5 An Information-Theoretic Account

We formalise why a coherence-maximising retriever can be paradox-prone. Consider a query Q, a candidate set $\mathcal{C} = \{c_1, \ldots, c_n\}$, and an unobserved ground-truth answer Y. Let $\mathrm{coh}(\mathcal{C})$ denote any monotone-increasing function of the off-diagonal pairwise similarity of the candidate set — CCS is one such function. Let $I(\mathcal{C}; Y \mid Q)$ denote the mutual information between the retrieved set and the answer, conditioned on the query.

Proposition 1 (Refinement Paradox)**.** *Let $\mathcal{C}_\star \subseteq \mathcal{C}$ be the subset that maximises $coh(\cdot)$ subject to $|\mathcal{C}_\star| = k'$ ($k' \leq |\mathcal{C}|$). If the per-passage relevance scores $\{r(c_i, Q)\}_i$ used by the refinement step are not perfectly aligned with each passage's marginal contribution to $I(\mathcal{C}; Y \mid Q)$, then there exist queries for which $I(\mathcal{C}_\star; Y \mid Q) < I(\mathcal{C}; Y \mid Q)$. In words: maximising coherence can strictly reduce the information content of the retained set.*

The proof is a one-line application of the data-processing inequality: the refinement step is a stochastic map $\mathcal{C} \to \mathcal{C}_\star$ that depends only on per-passage relevance, so any information uniquely supplied by a passage that is removed for being incoherent (with the rest) is lost — regardless of how relevant that passage was. The inequality is strict whenever such a passage exists.

Theorem 1 (Recovery Condition)**.** *A retriever $\mathcal{R}$ is paradox-free if it satisfies: $\forall \mathcal{C}$, $I(\mathcal{R}(\mathcal{C}); Y \mid Q) \geq I(\mathcal{C}; Y \mid Q)$. A sufficient condition is that $\mathcal{R}$ retain every passage whose marginal information contribution exceeds a fixed threshold, even when its per-passage similarity is low.*

Operationalisation. The HCPC-v2 rank-protection rule (§6) is one implementation of the recovery condition: by always retaining the top-k passages by raw query similarity, we preserve their information contribution even when refinement would otherwise discard them. The empirical recovery numbers in Table 1 are consistent with the bound: the gap $\mathrm{faith}_{\mathrm{base}} - \mathrm{faith}_{\mathrm{v1}}$ tracks the proportion of queries on which the protected passages would otherwise have been refined away.

The full information-theoretic derivation, including the dataset-level prediction (HotpotQA's multi-hop questions exhibit the largest paradox; SQuAD's single-span answers exhibit the smallest), the empirical bound verification (respected on 96% of $n = 150$ queries), and the connection to selection-bias literature, is in the longform appendix.

6 Context Coherence Score and HCPC-v2

This section formalises the diagnostic and the deployment policy. Both are deliberately small.

6.1 Context Coherence Score (CCS)

Let $\mathcal{C} = \{c_1, \ldots, c_k\}$ denote the top-k retrieved chunks for a query, and let $\mathbf{e}_i \in \mathbb{R}^d$ denote their L2-normalised embeddings. The Context Coherence Score is:

$$\mathrm{CCS}(\mathcal{C}) \;=\; \mu_{i<j}\big[\mathbf{e}_i \cdot \mathbf{e}_j\big] \;-\; \sigma_{i<j}\big[\mathbf{e}_i \cdot \mathbf{e}_j\big], \tag{1}$$

where the mean and standard deviation are taken over the $\binom{k}{2}$ off-diagonal pairwise cosine similarities. The subtraction term is load-bearing: a set with high mean similarity but high variance — two tight sub-clusters joined by no bridge — has the same mean as a set that is uniformly mid-similarity, and the latter is what the generator needs. CCS rewards the latter.

CCS is well-defined for any $k \geq 2$, requires only the embeddings already produced by the retrieval step, runs in $O(k^2 d)$ time, and is generator-free. For $k = 3$ (our default) it is $3d$ floating-point operations per query — negligible relative to the generator forward pass.

Decision policy. We treat CCS as a deployment-time gate. Given a similarity threshold $\tau \in [-1, 1]$, the bare CCS gate is:

```
docs, sims = retriever.retrieve(query)
ccs = compute_ccs(docs, embedder)
```

```
if ccs ≥ tau:
    return docs # coherent - pass to generator
else:
    return refine(docs) # incoherent - apply HCPC-v1
```

The bare CCS gate already recovers most of HCPC-v2's gain (§7, Table 2). We deploy it as the *decision policy*, and use HCPC-v2 as the implementation that adds two additional safeguards described next.

6.2 HCPC-v2: protected ranks + sub-chunk merge-back

HCPC-v2 extends the bare gate with two additional rules tuned on small held-out sets:

Rank-protected top-k. Even when the gate decides to refine, the top-2 retrieved chunks (by raw query similarity) are *never* touched. These are the chunks most likely to carry the answer span; refining them risks the exact failure mode the gate is meant to prevent.

Sub-chunk merge-back. Chunks that pass the gate-and-rank filter are subdivided into 256-token sub-chunks, each rescored by the cross-encoder, and only the top sub-chunk per parent is retained. The result is that refinement, when it happens, replaces a passage with its most discriminating fragment rather than a fragment from elsewhere in the corpus. We sweep the sub-chunk size in $\{128, 256, 512\}$ and report sensitivity in §7.

Hyperparameters. The coherence threshold $\tau = 0.50$ and the similarity / cross-encoder thresholds for the dual-AND gate ($\sigma_{\text{sim}} = 0.45$, $\sigma_{\text{ce}} = -0.20$) are tuned on a single held-out set of 50 SQuAD queries and held fixed across every experiment in the paper. We sweep $\tau \in \{0.4, 0.5, 0.6\}$ in the longform-paper appendix and confirm the policy is stable across a ± 0.1 window around the operating point.

Cost. Per query, HCPC-v2 adds: one CCS computation ($O(k^2 d)$ = a few thousand FLOPs); when the gate fires, $O(k)$ cross-encoder calls (~ 5 ms each on CPU); the optional sub-chunk pass adds $O(k \cdot \text{n_subs})$ further cross-encoder calls. Total marginal latency over the baseline pipeline: typically < 50 ms, dominated by the cross-encoder. For comparison, Self-RAG's reflection-token mechanism roughly doubles end-to-end latency.

Bare-gate ablation. A natural ablation isolates which of HCPC-v2's three moving parts is doing the work. We construct the bare CCS gate (no rank protection, no sub-chunks) and report its performance in the top-k ablation (Table 2, `gate rec.` column): on SQuAD it recovers 0.067 of HCPC-v2's 0.062 recovery at $k = 2$ — nearly identical. The gate decision is the active ingredient; rank protection contributes the bulk of the remaining gain only at larger k. We ship the bare gate as the recommended deployment policy (released as a `pip install context-coherence` package and a drop-in LangChain `CoherenceGatedRetriever`).

7 Robustness

The paradox could plausibly be an artifact of any of seven design choices we held fixed. We test all seven. The full per-experiment tables are in the longform-paper appendix; this section presents the distilled summary.

We highlight the four non-obvious entries.

Table 2: Seven robustness checks. Each row reports the headline $\Delta_{\text{paradox}} = \text{faith}_{\text{base}} - \text{faith}_{\text{v1}}$ and the HCPC-v2 recovery $\Delta_{\text{recovery}} = \text{faith}_{\text{v2}} - \text{faith}_{\text{v1}}$. The check is whether the result holds in the direction the coherence account predicts.

Axis	Variation tested	Headline finding	V
Multi-seed	3 random seeds	SQuAD: 0.069 ± 0.004 (17× signal/σ)	✓
Hierarchical (RAPTOR)	RAPTOR-2L vs HCPC-v2	HCPC wins 2/3 datasets, ties on hallucination	✓ co
Prompt template	strict / cot / concise / expert	ANOVA $p = 0.83$ (SQuAD); $p = 0.29$ (PubMedQA)	✓ no
Sub-chunk size	$\{128, 256, 512\}$ tokens	paradox visible in 3/3 cells per dataset	✓
Frontier scale	7B → 70B → 120B	SQuAD/70B exactly reproduces 7B magnitude (+0.100)	✓
Top-k	$k \in \{2, 3, 5, 10\}$	paradox *grows* with k on SQuAD (0.063 → 0.124)	✓ predi
Reranker family	MiniLM-L6/L12/BAAI-bge	paradox in all 6 cells; recovery in all 6	✓ rerar
Quantization	Q4/Q5/Q8	SQuAD paradox $\in [0.088, 0.095]$; PubMedQA ≈ 0 at all	✓ qua
Temperature	$T \in \{0, 0.3, 0.7, 1.0\}$	paradox *amplifies* with T on SQuAD (0.065 → 0.115)	✓ predi
Long-form generation	QASPER + MS-MARCO	paradox does not generalise; HCPC still cuts unsupported-rate −67%	† scop
Generic retrieval noise	inject $0/1/K$ off-topic	SQuAD noise slope > paradox magnitude	† qualitativ

Dataset	k	$\text{faith}_{\text{base}}$	faith_{v1}	Δ_{paradox}	gate rec.	v2 rec.
PubMedQA	2	0.599	0.553	+0.045	+0.019	+0.040
PubMedQA	3	0.604	0.566	+0.038	−0.018	+0.023
PubMedQA	5	0.586	0.561	+0.025	+0.003	+0.028
PubMedQA	10	0.585	0.586	−0.001	+0.004	−0.007
SQuAD	2	0.768	0.705	**+0.063**	+0.067	+0.062
SQuAD	3	0.790	0.721	**+0.069**	+0.071	+0.072
SQuAD	5	0.828	0.705	**+0.124**	+0.042	+0.139
SQuAD	10	0.814	0.697	**+0.117**	+0.018	+0.120

Top-k amplification. The coherence account predicts that increasing k adds relevance breadth (more matches) but spreads the retrieved set across more disparate sources, so coherence should fall and the paradox should grow. The full top-k table confirms exactly this on SQuAD: the paradox more than doubles from k=2 (+0.063) to k=5 (+0.124). The bare CCS gate's recovery (`gate rec.`) tracks HCPC-v2's recovery within ±0.05 at every k, supporting the claim that the gate decision is the active ingredient.

Frontier-scale persistence. The reviewer concern that the paradox is a small-model artifact is addressed by holding the retrieval stack fixed and swapping only the generator (Llama-3.3-70B and GPT-OSS-120B via Groq). On SQuAD, Llama-3.3-70B exactly reproduces the 7B-Mistral paradox magnitude (0.100 vs 0.100), and GPT-OSS-120B still shows a positive +0.030 paradox. PubMedQA paradox magnitudes shrink at scale (0.005/0.013 at 70B/120B vs 0.025 at 7B), which the coherence account predicts: stronger parametric priors are a more plausible substitute for fragmented biomedical evidence than for short-span SQuAD answers. The full table is in the longform-paper appendix.

Temperature amplification. A novel finding from the temperature sweep: $\text{faith}_{\text{base}} - \text{faith}_{\text{v1}}$ on SQuAD grows monotonically with T, from +0.065 at $T = 0$ to +0.115 at $T = 1.0$. The coherence interpretation is that higher T samples more aggressively from the parametric prior, and a fragmented context (HCPC-v1) gives the prior more room to operate. PubMedQA paradox magnitudes stay flat across T (consistent with the domain-mismatch ceiling story). HCPC-v2 recovery is robust across all T and is largest at $T = 0.7$ on SQuAD (+0.130).

Cross-validation: model self-confidence tracks CCS. We ask the same generator to rate its confidence (0–100) in the answer it just produced, given the retrieved context. Pooling SQuAD + PubMedQA ($n = 60$), self-confidence is significantly correlated with CCS: Pearson $r = 0.36$ ($p = 0.005$), Spearman $\rho = 0.48$ ($p = 0.0001$). The model's internal coherence-tracking signal aligns with the external diagnostic. Per-dataset correlations are weaker ($n = 30$ each), so the result is to be read as a pooled cross-validation rather than a strong per-dataset claim. Details in the longform-paper appendix.

clusters are tight under baseline, scatter under HCPC-v1, and recover under HCPC-v2.

Honest qualifications. Two robustness checks do *not* land where the strongest version of our claim would predict, and we report both transparently. First, the paradox does not generalise to long-form generation on QASPER or MS-MARCO; HCPC-v2 instead reduces the unsupported-claim rate (-67% on MS-MARCO). Second, on SQuAD the noise-injection slope exceeds the paradox magnitude, so we cannot claim the paradox is *magnitudinally larger* than generic retrieval noise — only that it is sign-distinct from noise on the multi-hop and biomedical regimes. Both qualifications are discussed in §8 and elaborated in the longform-paper appendix.

8 Discussion and Limitations

8.1 What CCS is — and is not

CCS is a *measurement*, not a remedy. It tells the deployer that a retrieved set is internally fragmented relative to what the generator needs. It does not tell the deployer how to fix it. In our deployment policy, the remediation is to *not* refine and trust the original retrieval. In other deployments the right move might be to re-retrieve with a larger k, fall back to zero-shot, or trigger external search. The diagnostic is decoupled from the response, by design.

8.2 Three regimes where CCS does not predict faithfulness

A theory is more credible when its proponents articulate where it breaks. We find three regimes in which CCS does not reliably predict the paradox:

(1) Highly redundant corpora. When the corpus is dominated by near-duplicate passages (legal boilerplate, software changelogs), every retrieval set has high CCS by construction. The signal saturates and provides no discrimination. Diagnostic: if the median CCS across queries exceeds 0.85, treat CCS as an availability check rather than a quality signal.

(2) Long-context generators with single-document inputs. If the generator's context window comfortably holds the entire relevant document and retrieval reduces to "select this one document", coherence is trivially 1.0 and the paradox does not arise — there is no fragmentation to begin with. This is consistent with our long-form non-result on QASPER (Table 2).

(3) Domain-mismatched encoders. When the encoder is general-purpose and the corpus is specialised (MiniLM on biomedical PubMedQA), every retrieval set has *low* CCS because the encoder cannot resolve fine-grained in-domain similarity. The CCS distribution collapses and the paradox is attenuated — but the failure mode is different: the issue is now upstream encoder mismatch rather than refinement-induced fragmentation. CCS still detects the regime (its distribution flags the encoder mismatch) but cannot fix it.

The common thread: CCS measures a specific structural property of the retrieved set. When that property is uninformative (saturated, trivial, or collapsed), CCS is uninformative. When it is informative, it is a clean predictor of refinement-induced failure.

8.3 Limitations

We mark four limitations explicitly:

Long-form generation. The paradox is a property of short-answer QA where the generator must extract a precise span. On long-form generation (QASPER, MS-MARCO multi-sentence) the same intervention reduces unsupported-claim rate but the paradox itself does not generalise in the form measured by single-span NLI.

Magnitude vs. generic noise. On SQuAD the noise sensitivity slope (-0.154) exceeds the paradox magnitude (0.100). We can claim the paradox is sign-distinct from generic retrieval noise on multi-hop and biomedical datasets, but we cannot claim its magnitude exceeds noise on SQuAD specifically. The original ratio ≥ 2 target was not met; we report the result honestly.

Adversarial set yield. The auto-generated 200-case adversarial set was filtered to 129 validated cases by an NLI validator + length/overlap heuristic. The mechanistic classifier in the appendix is trained on a 20-pair subset; full mechanistic validation across the 129-case set remains a follow-up.

Closed-model frontier. We evaluate up to GPT-OSS-120B via Groq's free tier; we do not evaluate 400B+ closed models (GPT-4, Claude-3.5). The coherence account predicts the paradox should attenuate further at very large scale on tasks where parametric memory can substitute for retrieval, and persist on tasks where retrieval supplies novel information. This is a clean prediction for follow-up work.

9 Conclusion

We have shown that better per-passage retrieval can make answer faithfulness *worse* when refinement fragments the coherence of the retrieved set. The phenomenon is robust across 5 datasets, 5 generator families (7B–120B), 9 ablation axes, and 7,000+ queries. We have introduced a generator-free retrieval-time signal (CCS) that detects the failure, validated it cross-signal against the model's own self-confidence, and shipped it as a deployment policy (HCPC-v2) that recovers faithfulness with 0% hallucination at one pairwise-similarity computation per query. The benchmark, the `pip install`-able metric, the LangChain integration, and a permanent Zenodo DOI are released.

The broader message is methodological. RAG evaluation has been optimising the wrong quantity. We have been measuring how good each passage looks on its own, when what the generator actually needs is evidence that fits together. CCS is a small step toward measuring the right thing.

Broader impact. RAG is increasingly deployed in high-stakes settings (medical QA, legal research, customer support) where unsupported claims have direct downstream cost. Our work identifies a counter-intuitive failure mode in the dominant deployment pattern (rerank + per-passage refinement) and provides a generator-free, deployment-time signal that flags it. The expected positive impact is on deployers who can now flag low-coherence retrieval sets before they reach the generator. A possible negative impact is mis-use of the diagnostic to over-confidently certify "high-coherence" outputs as faithful: we explicitly mark CCS as a *predictive signal, not a guarantee*, in §8.2 and identify three regimes where it does not predict.

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